

CS360 Introduction to Database

Course Information

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Basic Information (1)

- Course code: CS.30600
- Category: Mandatory / Core Elective / Major Elective
- Type: fully asynchronous online course (16 weeks)
 - You can watch the lectures at your preferred time and progress at your own pace
 - All lecture video links and slides are available on KLMS
 - The lecture videos can be viewed on YouTube
 - The videos were originally recorded in 2025
- The above theoretical lectures are essential for
 - SQL programming project
 - DBMS internal programming project
 - final exam

Basic information (2)

- In the era of Agentic AI, the course significantly **strengthens its hands-on programming projects** (called as the **Kite project**)
 - in addition to the theoretical lectures of 16 weeks
- No midterm exam, no homework, – but **three programming projects (60%)**, **a final exam (35%)** and **nano quizzes (5%)**
 - Programming projects: SQL (10%), RAG (25%), and DBMS internal (25%)
 - RAG: Retrieval-Augmented Generation (a technology that bridges AI and DB)
- **Students who cannot commit at least 12 hours per week are strongly advised not to enroll in this course**
 - Workload is particularly heavy from mid-April to mid-June
 - Students are strongly encouraged to watch the lecture videos in advance to better manage this period

Related (undergraduate) courses

- 200 level
 - CS204 Discrete Mathematics
 - CS206 Data Structure
- 300 level
 - CS300 Introduction to Algorithms (many algorithms in DBMS)
 - CS311 Computer Organization (DBMS exploits H/W features)
 - CS320 Programming Language (DBMS uses PL for interface)
 - CS330 Operating Systems and Lab. (DBMS resembles OS)

} actually, prerequisite

Course description

- Database systems serves as essential back-end systems across various industries
 - e.g., banking, e-commerce, retailer, manufacturing, health care, and modern AI applications
- This course is for understanding
 - what a database is
 - how to design a good database
 - what database queries are
 - how to write database queries
 - what a database management system (DBMS) is
 - how a DBMS processes queries

Part 1

Part 2

Course scope

- We will focus on the **de-facto standard and foundation**
 - relational data model
 - relational databases
 - relational DBMS (i.e., RDBMS)
 - relational queries (i.e., SQL)
- Hands-on programming projects
 - **SQL project** (for strengthen practical skills in using RDBMS)
 - **RAG project** (to develop skills in bridging LLMs, vector databases, and RDBMS)
 - **DBMS internal project** (to deepen understanding of RDBMS internals)
- A special lecture on vector DB and RAG will be provided in late April

Course schedule: Part 1

- Relational model (Week 1)
- Design theory for relational databases (Week 2-3)
- High-level database models (Week 4)
- Algebraic and logical query languages (Week 5)
- Database language SQL (Week 6-7)

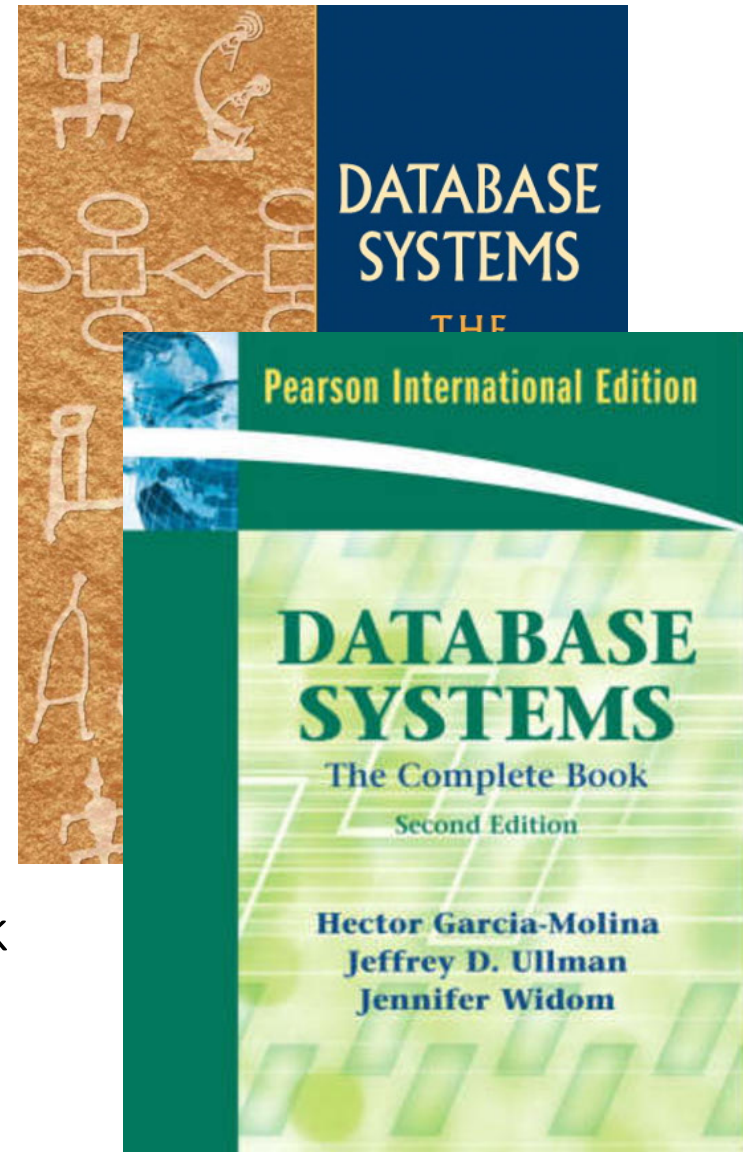
❖ Part 1 is related to the SQL project

Course schedule: Part 2

- Secondary storage management (Week 9)
 - Index structures (Week 10-11)
 - Query execution (Week 12-13)
 - Query compiler (Week 14-15)
- ❖ Part 2 is related to the DBMS internal project

Textbook

- Authors
 - Hector Garcia-Molina
 - Jeffrey D. Ullman
 - Jennifer Widom
- Edition: 2nd
- Publisher: Prentice Hall (2008)
- It is not necessary to purchase textbook
 - most of the course content is covered in the lecture slides



Reading assignments

- Week 1: ch. 1, ch. 2
- Week 2-3: ch. 3
- Week 4: ch. 4.1-4.6
- Week 5: ch. 5
- Week 6: ch. 6.1-6.4
- Week 7: ch. 6.5-6.6
- Week 9: ch. 13
- Week 10-11: ch. 14
- Week 12-13: ch. 15
- Week 14-15: ch. 16

Evaluation (1)

- Final exam (35%)

- written and OFFLINE exams in the auditorium
- scheduled between 6.15 and 6.19 (exact date TBD)
- duration: 3 hours

- Nano Quiz (5%)

- Designed to check your understanding while watching or after completing the lecture videos
- Provided in Google Form format
- Approximately 2~3 quizzes per class
- Cheating is strongly discouraged, as it will not help you prepare for the final exam

Evaluation (2)

- Kite project (Total 60%)

- KiteSQL (10%) – *Opens in early April*

- Focuses on mastering SQL queries, schema design, and database interaction using MySQL
 - You will act as a DBA managing a complex dataset

- KiteRAG (25%) – *Opens in mid-April*

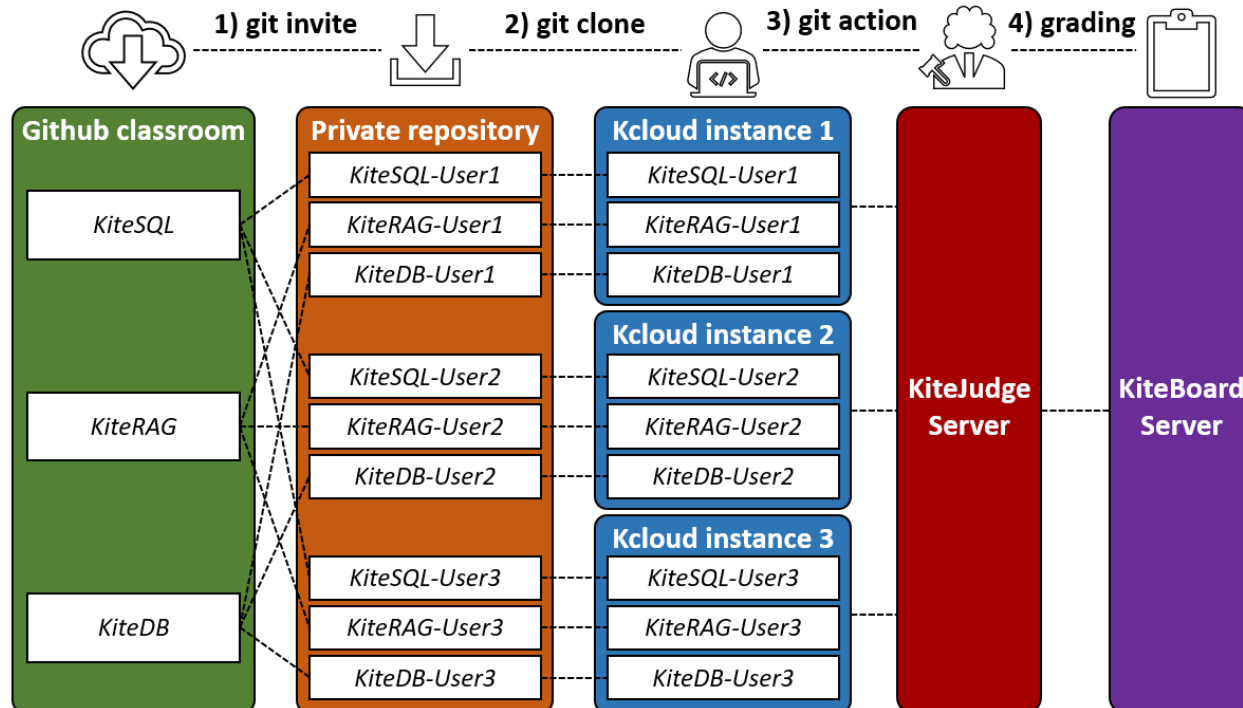
- Dive into the world of AI with Retrieval-Augmented Generation
 - You will build a system that retrieves relevant information to ground LLM responses, learning about vector databases and semantic search

- KiteDB (25%) – *Opens in early May*

- The capstone project where you implement core database internals
 - This involves understanding storage engines and query processing

Kite Project Workflow

- ① **Invitation:** Accept the assignment invitation via GitHub Classroom to get your private repository
- ② **Clone and Implement:** Clone your private repository to the Kcloud environment. Implement your code according to the instruction. You can test your code locally using the provided scripts
- ③ **Submit:** Push your code. GitHub Actions will automatically run the grading scripts in the KiteJudge server
- ④ **Grade and Compete:** Check your scores and rankings on KiteBoard and win awards



Nano Quizzes and Kite Project Schedule

Week	Lectures with <i>Nano quizzes</i> (5%)	Projects
Week 1 (3/2)	Relational model	No project
Week 2 (3/9)	Design Theory for Relational Databases	
Week 3 (3/16)		
Week 4 (3/23)		
Week 5 (3/30)	Algebraic and Logical Query Language	
Week 6 (4/6)	Database Language SQL	<i>KiteSQL (10%)</i>
Week 7 (4/13)		
Week 8 (4/20)	No midterm exam (additional RAG Lecture)	<i>KiteRAG (25%)</i>
Week 9 (4/27)	Secondary storage management	
Week 10 (5/4)	Index Structures	
Week 11 (5/11)		
Week 12 (5/18)	Query Execution	
Week 13 (5/25)		
Week 14 (6/1)	Query Optimization	No project
Week 15 (6/8)		
Week 16 (6/15)	<i>Final exam (35%)</i>	

Contact Information

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Thank you

Q & A